The notation used for double integral is $\int_a^b \int_c^d f(x,y)\,dy\,dx$

Here the function $f(x,y)$ is integrated with respect to y first and treat $f(x)$ constant and then integrate with respect to x and apply limits of x and simplify

Example 7.23

Evaluate: $\int_0^1 \int_1^2 x^2 y\,dx\,dy$, for $0 \le x \le 1,\ 1 \le y \le 2$

Solutions:

Let us first integrate with respect to y and then with respect to x. Hence the double integral is written as

$$\int_0^1 x^2 \left[\int_1^2 y\,dy\right] dx$$

Now integrate the inner integral only and simplify.

$$= \int_0^1 x^2 \left[\frac{y^2}{2}\right]_1^2 dx$$

$$= \int_0^1 \frac{x^2}{2}[4-1]\,dx$$

NOTE

We can also change the order of integration

Again integrate with respect to x

$$= \frac{3}{2}\int_0^1 x^2\,dx$$

$$= \frac{3}{2}\left[\frac{x^3}{3}\right]_0^1 = \frac{1}{2}[1^3-0]$$

$$= \frac{1}{2}$$

Example 7.24

Evaluate: $\int_0^2 \int_0^1 [2y^2x^2+3]\,dy\,dx$

Solution:

$$= \int_0^2 \int_0^1 [2y^2x^2+3]\,dy\,dx = \int_0^2 \left(\int_0^1 (2y^2x^2+3)\,dy\right) dx$$

$$= \int_0^2 \left[\frac{2x^2}{3}y^3+3y\right]_0^1 dx = \int_0^2 \left[2x^2\left(\frac{1}{3}\right)+3(1)-0\right] dx$$

$$= \left[\frac{2}{3}\left(\frac{x^3}{3}\right)+3x\right]_0^2 = \left[\frac{16}{9}+6\right]$$

$$= \frac{70}{9}$$